CRITICAL SOFTWARE X-RAY SPECIMEN

Payment succeeded. Access failed.

Make critical software safer to run, change and automate.

Synthetic professional-service specimen—not a client case study or production monitoring product.

- Synthetic demonstration based on the completed Rescue Lab incident.
- Not a real client case, testimonial or production incident report.
- Not live monitoring, legal advice, compliance assurance or financial assurance.
- Commercial scope and pricing are illustrative specimen values, not a quotation or guarantee.
- Reusable proprietary assessment internals are intentionally redacted.

X-RAY	ROADMAP
$2,250	$9,500

What failed, why it mattered, and what decision is required.

INCIDENT

Payment succeeded. Access failed.

WORKFLOW

Payment -> webhook -> queue -> subscription -> entitlement -> customer access

CONSEQUENCE

A synthetic paid subscription remained locked beyond its entitlement SLA.

FIRST DIVERGENCE

Queue/job provisioning failed three bounded attempts after payment and idempotent event processing succeeded.

DECISION REQUIRED

Contain paid-but-locked exposure and approve a controlled X-Ray before rewrite or automation.

CONFIDENCE

High for selected workflow; wider exposure unknown.

Confirmed impact: One synthetic account, USD 4,800 monthly subscription amount exposed while locked.

BUYER SHOULD UNDERSTAND

The cash path was healthy; customer value delivery was not.

BUYER SHOULD NOT INFER

This specimen does not prove real-client revenue loss or wider production exposure.

NEXT DECISION

Approve a scoped X-Ray before rewrite, broad retry or automation.

First proven divergence: Queue / job.

1	2	3	4	5	6	7
Payment HEALTHY	**Webhook** HEALTHY	**Event / processing layer** HEALTHY	**Queue / job** DIVERGED	**Subscription** WARNING	**Entitlement** DIVERGED	**Customer access** BLOCKED

EXPECTED

Entitlement provisioning succeeds within bounded attempts.

OBSERVED

3 failed attempts; deterministic timeout evidence retained.

RULE

Async fulfilment must either complete, expose failure or hand off to safe recovery.

RISK

This is the first proven divergence in the incident path.

EV-QUEUE

WHAT FAILED

Async fulfilment exhausted bounded attempts.

WHAT HELD

Payment and duplicate-delivery controls preserved one business effect.

WHAT TO AVOID

Do not blindly replay events or create replacement entitlements.

The issue is access-state divergence, not payment failure.

STAGE	EXPECTED	INCIDENT	RECOVERED	EVIDENCE
Payment	Succeeded payment remains authoritative	succeeded	succeeded	EV-PAYMENT
Webhook	Duplicate delivery retained without duplicate effect	10 deliveries / 1 event / 1 effect	No rewrite needed	EV-WEBHOOK
Queue	Provisioning completes or exposes bounded failure	3 failures	Handed to controlled reconciliation	EV-QUEUE
Subscription	Active paid subscription matches access	active subscription / locked access	active subscription / active access	EV-ENTITLEMENT
Entitlement	One active entitlement	one locked entitlement	one active entitlement	EV-RECOVERY
Remaining risk	Systemic exposure known	unknown outside fixture	still unknown outside fixture	EV-UNKNOWN

BEFORE

Paid + active subscription + locked entitlement

AFTER

Existing entitlement active once

REMAINING RISK

Wider exposure unknown outside the fixture

REJECTED EXPLANATION

Payment-provider failure is not supported by this evidence.

CONFIRMED REPAIR SHAPE

Update the existing entitlement, then verify repeat no-op.

OPEN QUESTION

A real X-Ray checks adjacent workflows and ownership boundaries.

Every material claim is tied to persisted evidence.

17:04:02	**Synthetic payment succeeded.** payment / succeeded	EV-PAYMENT
17:14:02	**10 signed deliveries retained for one event.** webhook / accepted	EV-WEBHOOK
17:14:02	**Provisioning attempt 1 failed.** queue / synthetic_provider_timeout	EV-QUEUE
17:14:02	**Provisioning attempt 2 failed.** queue / synthetic_provider_timeout	EV-QUEUE
17:14:02	**Provisioning attempt 3 failed.** queue / synthetic_provider_timeout	EV-QUEUE
17:14:03	**Entitlement existed but remained locked.** entitlement / active	EV-ENTITLEMENT
17:14:03	**Paid-but-locked SLA incident opened.** audit / recovered	EV-INCIDENT
17:14:04	**Approved action is bounded reconciliation of the existing entitlement.** approval / approved specimen decision	EV-RECOVERY
17:14:05	**Recovery activates one existing entitlement.** reconciliation / repaired / 1 effect	EV-RECOVERY
17:14:06	**Repeat recovery produces no_op and zero effects.** idempotency-check / no_op / 0 effects	EV-RECOVERY
17:14:07	**Verified outcome remains synthetic and scoped to the specimen workflow.** verification / verified in Lab	EV-STABILISED

OBSERVED

Payment, webhook, queue, entitlement and incident records.

VERIFIED

Recovery and repeat no-op are backed by command/test design.

UNKNOWN

Client-specific wider exposure is not inferred from the lab.

The entitlement provisioning job exhausted bounded synthetic timeouts.

UNDERLYING GAP

The paid-access path needed evidence-backed detection and controlled reconciliation after async fulfilment failure.

WHY NOT EARLIER

Payment and subscription state looked healthy until payment-to-access evidence was reconciled against entitlement state and SLA.

CONFIDENCE

High for the synthetic workflow; unknown beyond it.

UNKNOWNS

Whether the same control gap exists in other client workflows requires approved client evidence.

Competing explanations considered

EV-PAYMENT

Payment provider failure

Rejected for this fixture: payment is succeeded and authoritative.

EV-WEBHOOK

Duplicate webhook created duplicate customer effects

Rejected for this fixture: one event and one business effect were recorded.

EV-ENTITLEMENT

Access failed because no subscription existed

Rejected for this fixture: subscription became active while entitlement stayed locked.

Rules that must survive recovery, rewrite and automation.

RULE	STATEMENT	CONTROL	EVIDENCE
BR-01	Successful payment does not independently prove access.	Feature test for paid-but-locked detection.	EV-PAYMENT
BR-02	Repeated provider delivery must not create duplicate business effects.	Duplicate delivery feature test.	EV-WEBHOOK
BR-03	Active paid status requires a consistent entitlement.	Expected-vs-actual reconciliation assertion.	EV-ENTITLEMENT
BR-04	Stale or invalid events must not overwrite newer state.	Out-of-order and invalid event tests.	EV-WEBHOOK
BR-05	Recovery requires audit, idempotency and verification.	Repeat no-op recovery test.	EV-RECOVERY
BR-06	Rollback must not corrupt authoritative payment records.	Rollback checklist and audit review.	EV-PAYMENT

RULE	OWNER	FAILURE CONSEQUENCE	CONFIDENCE
BR-01	Engineering + Revenue Ops	Paid customer can remain locked without detection.	High
BR-02	Platform Engineering	Duplicate entitlements or repeated transitions.	High
BR-03	Product Engineering	Support tickets, churn risk and unsafe manual fixes.	High
BR-04	Platform Engineering	Access can be cancelled or mutated by unsafe replay.	High
BR-05	Engineering Lead	Manual repair can duplicate or hide the true incident.	High
BR-06	Engineering + Finance owner	Financial truth can be obscured by internal state rewrites.	Specimen inference

What must be fixed first.

CRITICAL / CONFIRMED DEFECT

F-01. Paid customer locked after successful payment

One synthetic paid account received no access beyond SLA.

Response:
Contain and reconcile affected paid subscriptions before broad changes.

Urgency:
Immediate

Acceptance:
Query finds zero paid active subscriptions with locked entitlements beyond SLA.

EV-PAYMENT / EV-ENTITLEMENT / EV-INCIDENT

HIGH / SYSTEMIC CONTROL GAP

F-02. Async fulfilment exhausted retries without closure

Retries were visible but did not complete access or trigger safe automatic closure.

Response:
Add bounded failure handling, alerting and operator handoff.

Urgency:
24-48 hours

Acceptance:
Exhausted jobs create actionable incident evidence.

EV-QUEUE

HIGH / CONFIRMED RISK CONTROLLED IN LAB

F-03. Manual repair needs repeat-safety proof

A duplicate repair path could create multiple entitlements or hide state divergence.

Response:
Use lock-protected reconciliation and run repeat no-op verification.

Urgency:
Before production repair

Acceptance:
Second repair returns no_op and entitlement count remains one.

EV-RECOVERY

MEDIUM / UNVERIFIED RISK

F-04. Wider exposure is unknown outside the selected path

Other paid workflows may share the same hidden business rules.

Response:
Scope a real X-Ray against one selected client workflow.

Urgency:
Before modernisation scope

Acceptance:
Approved evidence confirms or rejects similar exposure.

EV-UNKNOWN

MEDIUM / IMPROVEMENT OPPORTUNITY

F-05. Evidence standard should precede rewrite or AI automation

Unsafe automation can accelerate an unresolved state problem.

Response:
Modernise the critical path after evidence, tests and rollback are in place.

Urgency:
Before sprint planning

Acceptance:
Critical-path map, tests and rollback rehearsal accepted.

EV-STABILISED

Contain without creating duplicate effects.

ENGINEERING OWNER

Identify paid-but-locked accounts

Run scoped reconciliation query by workspace and SLA.

Validation: Sample accounts checked against payment and access state.

ENGINEERING LEAD

Freeze broad replay/retry actions

Pause bulk queue retries and manual entitlement creation.

Validation: No new duplicate entitlements appear.

APPLICATION ENGINEER

Run controlled recovery once

Execute transactionally scoped reconciliation.

Validation: Access active; entitlement count one.

APPLICATION ENGINEER

Verify repeat no-op

Run the same reconciliation again.

Validation: Second result is no_op; entitlement count remains one.

Do not do yet: Do not rewrite the billing system before evidence is mapped. Do not blindly replay every provider event or failed job. Do not create replacement entitlements without uniqueness and audit. Do not edit payment-provider history to make internal state appear consistent. Do not add an AI agent to a workflow whose business rules are still hidden.

Activate the existing locked entitlement through ReconciliationService.

PREREQUISITES

Confirm workspace, payment, subscription, entitlement and incident identifiers.

BOUNDARY

One workspace, one incident, existing entitlement only.

IDEMPOTENCY

Unique entitlement plus row locks; repeat action returns no_op and zero effects.

ROLLBACK TRIGGER

Unexpected workspace, entitlement count greater than one, repeat effect greater than zero, or mismatched payment/subscription evidence.

ROLLBACK ACTION

Stop further repair and apply forward correction after evidence review; do not rewrite payment history.

OWNER

Engineering lead with revenue owner approval.

VERIFICATION

- payment remains succeeded
- subscription remains active
- entitlement active
- access restored
- repaired action recorded
- repeat no_op recorded

MONITORING

Watch new paid-but-locked incidents, failed provisioning attempts and duplicate-effect counts.

Access is restored once; wider exposure is still not invented.

VERIFIED IN SYNTHETIC SCENARIO

Payment remains authoritative

EV-PAYMENT

VERIFIED IN SYNTHETIC SCENARIO

Subscription state is correct

EV-PAYMENT

VERIFIED BY COMMAND/TEST

Entitlement becomes active once

EV-RECOVERY

VERIFIED IN SYNTHETIC SCENARIO

Access is restored

EV-RECOVERY

VERIFIED BY IMMUTABLE ACTION MODEL/TEST

Recovery is audited

EV-RECOVERY

VERIFIED BY COMMAND/TEST

Repeat recovery is a no-op

EV-RECOVERY

NOT PRODUCTION PROVEN

Systemic risks remain visible

EV-UNKNOWN

Remaining risk: Similar workflows that collect payment before asynchronous fulfilment may share the same hidden rules. A real client X-Ray must confirm code paths, data quality, ownership, monitoring and release constraints.

$2,250 X-Ray first. $9,500 specimen sprint if justified.

DAYS 1-5

Containment and reconciliation

$2,500

Affected paid-access state is bounded and repaired safely.

Deliverables: paid-but-locked query; controlled recovery path; repeat no-op verification

Acceptance: Known affected records reconciled; no duplicate entitlements created.

Buyer: Provide approved access and name the workflow owner.

DAYS 6-12

Root repair and business-rule tests

$2,500

Hidden business rules are captured as executable controls.

Deliverables: Business Rule Atlas; feature tests; state-transition guardrails

Acceptance: Tests fail on paid-but-locked and duplicate-effect regressions.

Buyer: Confirm rule owners and SLA thresholds.

DAYS 13-21

Observability, idempotency and failure handling

$2,500

Retries, inbox state and recovery signals become operationally visible.

Deliverables: correlation trail; failure handoff; idempotency evidence

Acceptance: A replayed provider event creates one business effect and visible evidence.

Buyer: Approve operational alert thresholds.

DAYS 22-30

Controlled release, rollback rehearsal and handover

$2,000

The critical path can be changed without blind production risk.

Deliverables: release checklist; rollback rehearsal; engineering handover

Acceptance: Release and rollback steps are rehearsed with evidence captured.

Buyer: Schedule release window and responsible approver.

X-Ray fee may be credited if the sprint is signed within 14 days. Illustrative specimen pricing-not a quotation or guarantee.

What is included, excluded and required.

ASSUMPTIONS

- One selected critical workflow.
- Approved access to relevant code and sanitized evidence.
- No production credentials required for the first triage call.

EXCLUSIONS

- Full rewrite.
- Security/compliance certification.
- Financial audit.
- Broad codebase review outside selected path.
- AI copilot or autonomous agent build.

ACCESS REQUIRED

- Repository or architecture context.
- Sanitized events/logs when available.
- Engineering and business owner availability.

SCOPE-CHANGE CONDITIONS

- Multiple workflows.
- Missing evidence.
- Required infrastructure change.
- Security or compliance blockers.

NEXT DECISION

Approve the Critical Software X-Ray first; use its evidence to decide whether the 30-day sprint is responsible.

Change control: Production changes require buyer approval, release window and rollback owner. Buyer retains production deployment authority unless separately agreed.

An evidence-backed decision package, not a generic PDF.

- Critical-path map
- Evidence and state reconciliation
- Business Rule Atlas sample
- Prioritised risks
- Immediate containment decision
- Recovery and rollback plan
- Baseline and verification criteria
- Priced 30-day rescue roadmap
- Executive readout

Evidence register

EV-PAYMENT / OBSERVED

Payment confirmation

High

EV-WEBHOOK / OBSERVED

Signed webhook inbox

High

EV-QUEUE / OBSERVED

Provisioning attempts

High

EV-ENTITLEMENT / OBSERVED

Entitlement state

High

EV-INCIDENT / OBSERVED

SLA incident

High

EV-RECOVERY / VERIFIED

Recovery design

High

EV-STABILISED / OBSERVED

Stabilised replay

High

EV-UNKNOWN / UNKNOWN

Client-specific wider exposure

Explicit unknown

Request a 20-minute Critical Software Risk Triage

Bring one critical workflow that is failing, fragile or too risky to change. We will qualify the impact, urgency, evidence access and fit. The call is not a free architecture audit.

Request a 20-minute Critical Software Risk Triage

CTA destination remains configurable through TRIAGE_CTA_URL.

Bring one critical workflow:
failing, fragile or too risky to change.